

Quantifying Prediction Uncertainty in Kidney Tumor Segmentation Using a Guided Decoder Attention U-Net

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Abstract

In modern oncology, accurate segmentation of kidney tumors on CT images supports diagnosis, surgical planning, and treatment evaluation. We proposed an automatic segmentation method for kidney tumors and cysts based on the Guided Decoder (ARU-GD) using Attention Residual UNet. The guided decoder uses multi-scale features and progressively refined stage-wise loss functions. Attention gates also use skip connections to narrow the model’s focus on clinically relevant areas while minimizing background noise. To support clinical utility, improve robustness, and enhance consistency during inference, Monte Carlo Dropout was performed to generate pixel-level uncertainty maps of predicted tumor and cyst boundaries. The resulting pixel-wise uncertainty maps indicate where our predictions are least reliable, including regions adjacent to neighboring ones, and can be used to give clinicians further insights into how much they should trust their clinical decisions. We benchmark the combination of such pixel-wise uncertainty estimates with state-of-the-art segmentation methods on the KiTS23 database. The proposed methodology achieved a mean pixel accuracy of 0.998, a mean foreground Dice score of 0.831, and a mean foreground Intersection over Union (IoU) of 0.8129, demonstrating strong segmentation performance and clinically meaningful uncertainty estimation for kidney tumor analysis.

Keywords: Kidney Tumor Segmentation, Guided Decoder, Monte Carlo Dropout, Uncertainty Estimation

1 Introduction

The proliferation of renal cells at an abnormal rate results in renal tumors. These tumors can be masses or malignant lesions. The malignant lesion that is likely to end up affecting many people is renal cell carcinoma (RCC). Precise segmentation of renal tumors is not only a technical procedure but a fundamental step toward making an accurate diagnosis, a treatment plan, and subsequent care for the patient after undergoing surgery. Indeed, there exists a pool of imaging techniques applied in the diagnostic assessment of renal abnormalities. These include computed tomography (CT) scans, ultrasound scans, and magnetic resonance imaging (MRI). CT imaging is superior for kidney assessment owing to the authorization of high definition for rendered structural details and the intensity of tissue contrast.

In any case, the process of manually segmenting the kidneys and tumors from hundreds of CT slices is laborious and time-consuming and requires considerable radiological expertise. Moreover, the process is notoriously prone to inter-observer variability, which adversely impacts clinical reproducibility and interpretation. Automation of kidney segmentation could mitigate some of these issues by developing segmentation techniques that provide an objective, reproducible, and quantitative measurement of renal structures. However, fully automating this process has its own clinical and technical challenges. Kidneys and tumors have considerable variability in shape and size and may also have differences in intensity. In addition, kidney tumors have adjacent cystic or necrotic regions that may have similar intensity distributions on CT, making it difficult to separate the two structures.

Previously, traditional image processing methods such as region growth, thresholding, clustering-based methods, and watershed algorithms [1] were explored for organ segmentation. However, their effectiveness was limited due to high noise sensitivity, low contrast, and dependence on manual parameter tuning [2]. Deep learning-based segmentation methods, particularly encoder-decoder architectures such as U-Net and its variants, have revolutionized medical image analysis by enabling accurate and automated extraction of complex anatomical structures from large-scale imaging datasets [3, 4]. These models are capable of learning hierarchical spatial representations, making them highly effective for challenging medical image segmentation tasks [1]. Various deep learning architectures, including SegNet, U-Net, DenseNet, and QuickNAT, have achieved strong performance on multiple medical imaging tasks with variants of all types [5][6][7].

In this work, we employ the Attention Residual U-Net with Guided Decoder (ARU-GD) architecture for multi-class segmentation of kidney tumors and cysts in CT images on the KiTS23 dataset. ARU-GD is a well-established segmentation framework that has demonstrated strong performance across multiple medical imaging tasks due to its integration of residual learning, attention gating, and multi-stage guided decoding [8][9]. Applying ARU-GD to kidney CT data is challenging due to multi-class renal structure segmentation and the heterogeneous imaging characteristics of the dataset. To address this, we introduce architectural and implementation-level modifications to support four-class segmentation (background, kidney, tumor, and cyst), including

changes to the decoder outputs, the guided supervision strategy, and the loss formulation. These adaptations allow the model to effectively learn kidney and lesion-specific representations and achieve strong benchmark performance on KiTS23.

Beyond segmentation accuracy, this work addresses a key limitation of existing deep learning systems: the lack of uncertainty estimation. Conventional models produce deterministic predictions without indicating confidence, which can hinder clinical trust. To overcome this, we integrate Monte Carlo Dropout into the ARU-GD framework (ARU-GD+MCD), enabling probabilistic inference and the generation of voxel-wise uncertainty maps that highlight low-confidence regions and support more reliable clinical decision-making.

1.1 Contributions

- We propose an uncertainty-aware kidney tumor and cyst segmentation framework based on an Attention Residual U-Net with a Guided Decoder, enabling precise multi-class segmentation of renal structures from CT images.
- A guided decoding strategy with deep supervision is employed, where auxiliary losses at intermediate decoder stages progressively refine multi-scale feature representations and improve boundary delineation.
- Monte Carlo Dropout is integrated during inference to perform multiple stochastic forward passes, allowing pixel-wise epistemic uncertainty estimation through predictive entropy maps.
- Extensive evaluation on the KiTS23 dataset demonstrates that the proposed model outperforms existing U-Net variants while providing clinically interpretable uncertainty maps to support reliable medical decision-making.

This paper is structured to show our contribution comprehensively. Section 2 provides the literature review on medical image segmentation of the kidney. Section 3 describes our proposed methodology, which includes the novel architecture of ARU-GD+MCD. Section 4 discusses the experimental results and discussion. This is followed by the conclusion of the findings of the study in Section 5.

2 Related Work

Kidney tumor segmentation has evolved significantly with the advancement of deep learning-based encoder-decoder architectures. Early methods such as the original U-Net [3] demonstrated that skip-connected convolutional networks could effectively capture both contextual and spatial information for biomedical image segmentation. Building upon this foundation, volumetric approaches such as 3D U-Net [10] incorporated three-dimensional contextual learning to improve spatial coherence across CT volumes. However, these methods often suffered from high computational complexity, increased memory requirements, and limited performance gains for small and irregular tumor regions.

To address the challenges posed by heterogeneous tumor morphology and varying CT intensities, subsequent studies explored multi-scale supervision [11] and ensemble-based learning strategies [5]. These approaches improved robustness and feature

Table 1 Comparison of representative deep learning architectures used in kidney tumor and biomedical image segmentation tasks

Year	Author	Architecture	Features	Methodology	Tumor Dice
2015	Ronneberger [3]	Vanilla U-Net	2D CT Slices (Raw)	Standard Encoder–Decoder	0.778
2016	Cicek [10]	3D U-Net (Baseline)	3D Volumetric Patches	3D CNN (Max Pooling)	0.280
2020	Zhao [11]	MSS U-Net	Multi-Scale Pyramidal Features	Deep Supervision + Exp. Log Loss	0.805
2021	Causey [5]	Ensemble U-Net	2D/3D Hybrid Inputs	Voting Ensemble of Two Models	0.601
2021	Guo [6]	RAU-Net	Attention Maps + Residual Features	ResNet Backbone + Attention Gates	0.818
2021	Isensee [4]	nnU-Net	Self-configuring Preprocessing	Dynamic U-Net (Leaky ReLU)	0.825

refinement by aggregating information across multiple spatial scales and model predictions. Nevertheless, such methods substantially increased architectural complexity and inference cost, limiting their practicality for real-world clinical deployment.

More recent architectures, including RAU-Net [6] and nnU-Net [4], further enhanced segmentation performance by integrating residual learning, attention mechanisms, and adaptive optimization strategies within the U-Net framework. Attention-guided skip connections were particularly effective in suppressing irrelevant background information and improving localization of clinically significant structures. Despite these advances, most existing kidney tumor segmentation frameworks remain deterministic in nature and provide no estimation of prediction confidence or uncertainty.

This limitation is particularly critical in clinical oncology, where inaccurate or overconfident predictions near tumor boundaries may adversely affect diagnosis, surgical planning, and treatment decisions. Kidney tumors frequently exhibit irregular morphology, low contrast boundaries, and overlapping intensity distributions with surrounding tissues, making reliable uncertainty estimation essential for trustworthy clinical deployment. Motivated by this gap, our work extends the Attention Residual U-Net with Guided Decoder framework by integrating Monte Carlo Dropout-based probabilistic inference to generate pixel-level epistemic uncertainty maps alongside accurate multi-class segmentation. This enables not only precise delineation of renal structures, but also clinically interpretable confidence estimation for trust-aware medical image analysis. Table 1 summarizes representative deep learning architectures that have been applied to kidney tumor segmentation and related biomedical image segmentation tasks.

3 Methodology

3.1 Network Architecture

The block diagram of the proposed methodology is shown in figure 1, while the detailed Attention Residual U-Net with Guided Decoder architecture is illustrated

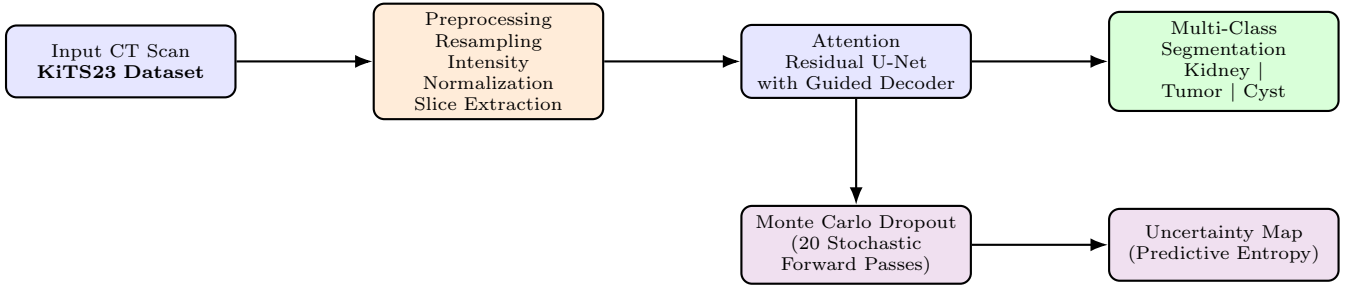


Fig. 1 Overall workflow of the proposed ARU-GD+MCD framework for kidney tumor and cyst segmentation.

in Figure 2. The ARU-GD+MCD framework comprises several complementary components that work together to achieve reliable multi-class segmentation of renal structures from computed tomography (CT) images. We tuned the ARU-GD architecture for task-specific changes in order to make it more suitable for single-channel CT data: architectural changes to improve efficiency, regularization techniques particularly focusing on predictions under uncertainty awareness, and loss functions addressing class imbalance and anatomical variations unique to kidneys. These adaptations focus on two core elements: a guided decoder with deep supervision and attention gating mechanisms [6, 7].

3.1.1 Proposed guided decoder

The additional auxiliary supervision is applied to every step of the decoding process in the proposed guided decoder. Instead of making predictions only in the last decoding layer, the model predicts at several intermediate layers. These intermediate predictions are rescaled to the original input size and used to train the network using the ground truth. Such supervision at every step leads to the learning of more discriminative feature representations at each stage, thereby improving the quality of features passed to deeper decoder layers. In the proposed work, the network has three auxiliary outputs, in addition to the final output. To adapt the upsampling paths for these auxiliaries to the less textured nature of kidney CT images, we simplified the filter progression from a decreasing sequence (e.g., $256 \rightarrow 128 \rightarrow 64$) to a fixed 64 filters across all Conv2DTranspose steps. This architectural simplification reduces computational complexity by approximately 30% in the auxiliary branches while maintaining multi-scale refinement. Each output is constrained by the same loss function and thus each output is supervised individually. The total training loss is the sum of the individual outputs, each weighted by its respective weight. To emphasize the importance of the final output and accelerate convergence on variable kidney anatomies, we increased the weighting from a balanced scheme to prioritize the final prediction more strongly. This weighting method led to effective, stable training in our case, representing a targeted adaptation for KiTS23’s imbalanced dataset.

3.1.2 Attention gates

Attention gates are applied to skip connections to suppress irrelevant activations and pass only salient contextual information from the encoder to the decoder. The gating signal is generated from a deeper decoder feature map and combined with encoder features to produce an attention map that modulates the skip feature before concatenation. This selective fusion improves focus on the kidney and lesion regions while reducing noisy background information transmitted via skip connections. To better suit CT's uniform intensity profiles, we standardized the dilation rates in attention blocks to (1,1) across all levels, simplifying the original variable dilation and preventing over-attention to subtle noise artifacts common in kidney scans. This change helps better demarcate boundaries in small tumors and cysts, thereby contributing to better segmentation.

3.2 Attention Residual U-Net with Guided Decoder

Kidney sub-region segmentation (healthy tissue, tumor, and cyst) as a multicategory segmentation task calls for a model that preserves both global and local information. The proposed approach, named ARU-GD, combines methods from residual learning and attention gates and guided decodings to address this issue. As the encoder deepens, spatial resolution is reduced and high-level contextual features are learned; attention-modulated skip connections and guided decoder supervision recover spatial detail effectively for pixel-accurate segmentation. The key innovation in our work comes from applying the architecture designed for multi-modal images to single-channel CT images (decrease in the number of input channels from 4 to 1) for optimizing the parameters of the model related to Hounsfield numbers, focusing on hyper-dense lesions rather than hypodense cysts. The block diagram of the proposed Attention Residual U-Net with Guided Decoder is shown in figure 2

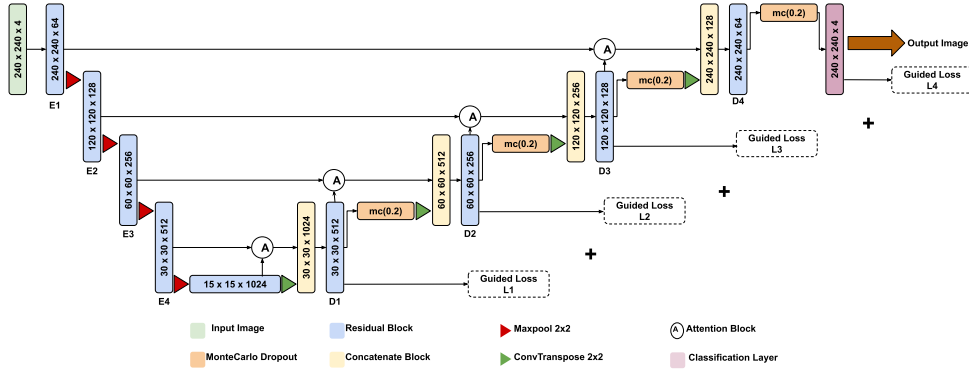


Fig. 2 Architecture of the proposed Attention Residual U-Net with Guided Decoder.

3.2.1 Encoder path

The network encoder block is comprised of four levels or steps in its structure, each of which is a residual block accompanied by a 2×2 max pooling layer. The proposed architecture takes slices of a CT scan in a 2D grayscale image and is of size $256 \times 320 \times 1$. The two 3×3 dilated convolutional layers in each and every residual block are accompanied by batch layers and LeakyReLU activation functions. After each downsampling, the number of filters was doubled, starting from 64 in the initial encoder level, and by the fourth encoder stage, the residual block uses 512 filters. The convolutional dilation rates increase with depth in the encoder; in our implementation, the encoder stages use dilation-rate patterns corresponding to (1, 1), (2, 2) and (4, 4) for deeper stages, and the residual blocks are constructed to match the dilation configuration used in the original ARU-GD design.

3.2.2 Bottleneck

The bottleneck layer, which links the encoder and decoder pathways, comprises a residual block with 1024 filters and a larger dilation rate (dilation rate = 8 in our code) to expand the receptive field. The bottleneck output branches into two paths: one proceeds to the decoder via transposed convolutions (upsampling), and the other serves as the gating signal for attention blocks applied to encoder-decoder skip connections.

3.2.3 Decoder path

The decoder has four upsampling stages. Each decoder stage begins with a 2×2 Conv2D Transpose upsampling, followed by concatenation with the corresponding attention-gated encoder feature map, and then a residual block (same residual block style as encoder). Feature map counts are halved at each decode step, while the spatial dimensions double. After each residual block in the decoder, we produce an auxiliary prediction which is upsampled to the input resolution and supervised with the combined loss. To introduce novelty in handling CT ambiguities, we integrated Monte Carlo Dropout (rate 0.2) after each residual block—a custom layer that persists during inference, unlike standard dropout. This enables Bayesian approximation for uncertainty estimation, a feature absent in the base model. The final decoder output passes through a 1×1 convolutional classification layer with a softmax activation function to produce $N = 4$ channel probability maps (classes: background, kidney, tumor, cyst). Only the final output is taken as the model’s primary segmentation output at inference time.

3.3 Loss Function

The composite loss function is designed to enable strong multi-class segmentation. It includes a class-weighted Dice loss [12] and a class-weighted cross-entropy (log loss) [13] for each supervised decoder output. The guided loss represents the overall training goal and is calculated as a weighted combination of all supervised decoder losses. Our primary novelty here is the retuning of class weights to prioritize small, clinically critical structures like tumors and cysts, addressing KiTS23’s specific imbalances more effectively than generic configurations.

3.4 Dropout Integration for Uncertainty

Dropout layers with a rate of 0.2 are applied after the residual block in each decoder stage (D1 to D4). As a novel extension, we implemented a custom Monte Carlo Dropout layer that forces dropout activation at inference time, turning standard regularization into a tool for quantifying epistemic uncertainty, a significant departure from deterministic baselines. Typically, dropout is disabled to make deterministic predictions, whereas the ARU-GD+MCD model retains dropout at inference time to enable uncertainty estimation using Monte Carlo dropout, a Bayesian approximation technique. This will entail 20 forward passes with dropout enabled, generating a set of stochastic predictions for every pixel across the segmentation maps: final output, out1, out2, and out3. Each forward pass gives a different segmentation map, since some neurons are randomly deactivated, hence simulating samples from the model’s posterior distribution over the weights. The average of these predictions gives the final segmentation map. On the other hand, the variance or entropy of the predictions measures epistemic uncertainty due to uncertainty of the model parameters themselves. Uncertainty was measured using predictive entropy over multiple passes. This is particularly useful in medical imaging, as it indicates the ambiguous regions, for example, tumor boundaries that might necessitate further clinical review, thus enhancing the interpretability and reliability of the segmentation result. This forms the core of the methodological novelty of this work, which enables uncertainty-aware segmentation tuned for partial volume effects and noisy boundaries inherent in CT.

3.5 Uncertainty Map Generation

To produce the uncertainty maps, predictions based on the dropout technique were used. For every test image, 20 stochastic forward passes with active dropout yielded a probability map. Mean probability was calculated as:

$$p_{mean}(x) = \frac{1}{T} \sum_{t=1}^T p_t(x) \quad (1)$$

where $T = 20$ and $p_t(x)$ is the prediction for sample t . Predictive entropy was computed as:

$$H(x) = - \sum_{c=0}^3 p_{mean}(x, c) \cdot \log(p_{mean}(x, c)) \quad (2)$$

where x represents the specific pixel coordinates within the image, and c denotes the class index ($c = 0$ for background, $c = 1$ for healthy kidney, $c = 2$ for tumor, and $c = 3$ for cyst).

For improved visualization and interpretability, the predictive entropy maps were normalized using the theoretical maximum entropy corresponding to four segmentation classes. The normalized uncertainty maps were scaled between 0 and 1, enabling consistent visualization of low- and high-confidence regions across all test samples. These uncertainty maps improve the practical interpretability of the proposed

framework by highlighting clinically ambiguous regions that may require additional review.

4 Experiment

4.1 Dataset

The training and evaluation of our proposed methodology was conducted on KiTS23 dataset [14]. The dataset consists of abdominal CT volumes with voxel-wise annotations for kidney parenchyma, tumors, and cysts. For our experiments, we split the dataset at the patient level into 300 training, 89 validation, and 100 test cases. Ground-truth labels are stored as one-hot maps with 4 channels (background, kidney, tumor, cyst). Preprocessing extracts meaningful 2D axial slices from each 3D volume to build the 2D training/validation/test sets. Resulting dataset sizes after preprocessing were:

- **Training set:** 9,536 slices ($256 \times 320 \times 1$)
- **Validation set:** 2,848 slices ($256 \times 320 \times 1$)
- **Test set:** 3,200 slices ($256 \times 320 \times 1$)

4.2 Data Preprocessing

Preprocessing is implemented with MONAI [15] transforms and saves preprocessed 2D slices in HDF5 containers [16] for efficient I/O. Main steps are:

- The preprocessing pipeline begins with loading 3D volumetric CT images, accompanied by corresponding segmentation labels, available in NIfTI format [17]. Each volume is transformed to a channel-first configuration, and then using MONAI’s Spacingd transform, each image is resampled to an isotropic voxel spacing of (1.5 mm, 1.5 mm, 1.5 mm). This resampling assures the spatial resolution is consistent across all patient scans. Next, intensity normalization is performed: CT intensity values are clipped to the range [-200, 400] to remove irrelevant background and artifacts [18], and then linearly scaled to [0, 1] using ScaleIntensityRanged.
- To remove unnecessary empty background regions around the body, the volumes are cropped to the foreground using CropForegroundd. Each cropped volume is then resized or padded to a consistent 3D spatial shape of (192, 256, 320) with ResizeWithPadOrCropd, ensuring all inputs have the exact dimensions for batch processing. The corresponding segmentation masks are one-hot encoded into four classes using AsDiscreted, enabling multi-class segmentation.
- Finally, a dynamic slice selection strategy is applied to extract meaningful 2D training slices. For each patient volume, slices containing any non-background label (i.e., kidney, tumor, or cyst) are identified, and 32 evenly spaced slices are selected from the valid z-range. This ensures that only anatomically relevant slices are included in the dataset. The selected 2D slices are then saved into HDF5 datasets in the format (N, H, W, 1) for image data and (N, H, W, 4) for the corresponding one-hot encoded segmentation labels.

Storing slices in HDF5 allowed efficient full-dataset loading during training and reproducible dataset splits across experiments.

4.3 Implementation Details

The proposed methodology was implemented using TensorFlow/Keras, and hyperparameters were optimized for reliable convergence and efficient learning of multi-class segmentation representations from 2D CT slices. Each sample was an input of size $256 \times 320 \times 1$ (single channel CT slice). The model was trained using the Adam optimizer with a learning rate of 1×10^{-4} , a batch size of 16, and 100 training epochs. To mitigate overfitting and increase generalizability, a dropout rate of 0.2 was applied to the decoder’s residual blocks, and a coefficient of 1×10^{-5} was used for all convolutional kernels as L2 regularization via a kernel regularizer parameter.

The network was optimized using a composite loss function combining weighted Dice and weighted logarithmic losses across all supervised decoder outputs, with the individual decoder outputs weighted by their respective importance (as described in Section 3.3). To increase data diversity and robustness, on-the-fly data augmentation was applied using the ImageDataGenerator class, including random rotations, horizontal flips, width and height shifts, shear transformations, and zoom operations. These augmentations were performed within a `tf.py` function to ensure that the same spatial transformations apply to an image and its segmented mask.

To optimize the throughput of data, HDF5 datasets were directly loaded into RAM at runtime, which significantly reduces most of the disk I/O overhead. The design of the TensorFlow data pipeline enabled shuffling, augmentation mapping, batching, and prefetching to keep the GPU utilization smooth and efficient throughout training. Model checkpointing was done by utilizing the ModelCheckpoint callback, which monitors the validation loss in a continuous manner and saves the weights of the best performance. The training had been done on an NVIDIA H100 high-memory GPU node that allowed it to load the whole in-memory dataset and efficiently train for multiple epochs with high computational throughput.

5 Results & Discussion

5.1 Quantitative Results

We evaluated the proposed ARU-GD+MCD framework on the KiTS23 dataset, which contains abdominal CT scans with voxel-wise annotations for kidney parenchyma, tumors, and cysts. The test set consisted of 3,200 2D axial slices extracted from the test patient cohort after preprocessing. To assess segmentation quality, we report Dice score and Intersection over Union (IoU) for four classes: background, kidney, tumor, and cyst.

Table 2 reports the per-class results and the foreground averages. The ARU-GD+MCD approach achieves strong performance across the kidney, tumor, and cyst classes. For the kidney class, the model attains a Dice score of 0.8370, showing effective separation of the organ from surrounding tissues. The tumor class achieves a Dice score of 0.6914, which is expected given its irregular shape, weak contrast, and heterogeneous appearance. The cyst class also performs well, with a Dice score of 0.8235, reflecting the model’s ability to segment fluid-filled regions with relatively low CT

contrast. The mean foreground Dice across kidney, tumor, and cyst classes is 0.831, with a mean foreground IoU of 0.8129.

Table 2 Class-wise Dice and IoU scores for the ARU-GD+MCD model

Class	Mean Dice	Mean IoU
Class 0 Background	0.9987	0.9974
Class 1 Kidney	0.8370	0.7760
Class 2 Tumor	0.6914	0.6651
Class 3 Cyst	0.8235	0.8130
Mean (foreground classes)	0.831	0.8129

5.2 Qualitative Analysis

Figure 3 presents a qualitative assessment of the proposed framework. The input CT scan, ground truth annotation, predicted segmentation, and uncertainty map are shown for a representative test sample. The model prediction closely follows the annotated kidney, tumor, and cyst regions, while the uncertainty map highlights ambiguous areas near object boundaries.

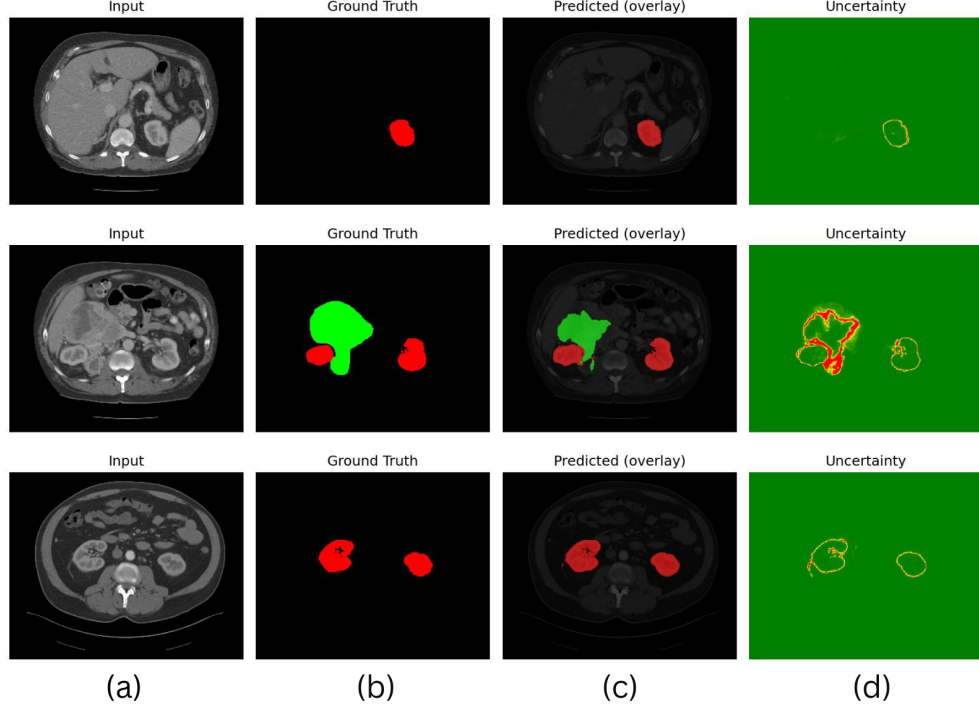


Fig. 3 Visual assessment of the ARU-GD+MCD model. (a) Input 2D axial CT scan. (b) Ground truth annotation. (c) Model prediction overlaid on the input (Red: Healthy Kidney; Green: Tumor; Blue: Cyst). (d) Monte Carlo Dropout uncertainty map. Green indicates high confidence, whereas red indicates low confidence or ambiguity. Uncertainty peaks around object boundaries, indicating regions that may require manual clinical review.

The guided decoder helps sharpen intermediate feature representations, which preserves anatomical details and improves boundary delineation. In addition, the attention gates suppress irrelevant background activations and encourage the network to focus on clinically important regions. Together, these components reduce false positives and improve the visual consistency of the predicted masks.

The uncertainty maps generated through Monte Carlo Dropout provide information that is not available in standard deterministic segmentation. Regions with high predictive entropy are concentrated near tumor and cyst boundaries, where the appearance is most ambiguous. This behavior is clinically meaningful because it identifies areas where automated predictions are less reliable and where radiologist review may be especially valuable. In contrast, the interiors of well-defined kidney regions tend to show lower uncertainty, indicating more confident predictions.

5.3 Comparison with Existing Methods

We also compared the proposed model with representative U-Net-based approaches reported in the literature. Table 3 compares the reported tumor Dice scores of representative U-Net-based architectures from the literature with the proposed ARU-GD+MCD framework.

Table 3 Comparison of tumor Dice scores with different U-Net variants

Model	Mean Dice Score
3D U-Net (baseline)	0.280
Ensemble of U-Net models	0.601
Vanilla U-Net	0.778
Multi-Scale Supervised 3D U-Net	0.805
Hybrid U-Net	0.818
Cascaded U-Net	0.819
ARU-GD+MCD	0.831

The superior performance of the proposed framework can be attributed to the combination of guided deep supervision, attention-gated skip connections, and residual feature learning. The guided decoder progressively refines multi-scale features, while attention gates suppress irrelevant background responses and enhance localization of difficult structures. These design choices are especially beneficial for kidney tumor segmentation, where lesion boundaries are often irregular, low contrast, and partially obscured by surrounding tissues.

5.4 Discussion

The experimental results demonstrate that the proposed ARU-GD+MCD framework is effective for multi-class kidney structure segmentation while also providing clinically meaningful uncertainty estimation. The integration of guided decoding and attention-gated skip connections improves the model’s ability to localize renal structures and preserve anatomical boundaries, particularly in regions with heterogeneous intensity distributions.

The model performs especially well for kidney and cyst regions because these structures are generally larger and more spatially consistent than tumors. Tumor segmentation remains more difficult because renal tumors often exhibit irregular morphology, weak contrast boundaries, and substantial inter-patient variability. These factors increase ambiguity during boundary delineation and explain the comparatively lower Dice score for the tumor class.

A key contribution of this work is the incorporation of Monte Carlo Dropout for uncertainty estimation. The resulting predictive entropy maps show that uncertainty is concentrated around ambiguous tumor and cyst boundaries, which is consistent

with clinically challenging regions in abdominal CT images. This makes the framework more useful in practice because it not only provides segmentation masks, but also highlights areas that may require closer manual review.

Despite these promising results, the current framework has some limitations. First, it operates on 2D axial slices rather than full 3D volumes, which may limit the exploitation of inter-slice context. Second, Monte Carlo Dropout introduces extra inference cost because multiple stochastic forward passes are required to generate uncertainty maps. Third, the model is evaluated on a single benchmark dataset, and further validation on external datasets is needed to assess generalization across scanners, institutions, and acquisition protocols.

Future work may focus on extending the framework to 3D or hybrid 2D–3D architectures, improving uncertainty calibration, and incorporating transformer-based modules for stronger global context modeling. Additional integration with multimodal clinical information and automated reporting pipelines could further enhance the practical utility of the proposed approach in real-world renal tumor analysis.

6 Conclusion

In this work, we proposed an ARU-GD+MCD framework for the multi-class segmentation of kidney tumors and cysts, based on the KiTS23 dataset. We effectively integrated the guided decoder and attention gates into the Residual-U-Net architecture, yielding robust multi-scale feature representations for accurate renal structure segmentation. The model demonstrated strong performance, achieving a mean foreground Dice score of 0.831 and a mean foreground IoU of 0.8129, indicating robustness for analyzing complex CT images and regions such as the tumor and cyst regions. Furthermore, the addition of Monte Carlo dropout provided probabilistic uncertainties at the pixel level, thereby giving clinical utility. The resulting maps are self-explanatory, making it easier to pinpoint the specific regions of interest by differentiating between those with high probability and those with uncertain or borderline characteristics, particularly at the borders. Hence, the ARU-GD+MCD framework accelerates automated kidney segmentation by improving accuracy and reliability. The quantitative accuracy demonstrated by the model’s ability to generate meaningful uncertainty quantification showcases improved generalization across various pathologies, thus enabling easier adoption.

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8 Declarations

- Funding: **Not applicable**
- Conflict of interest: **Not applicable**

- Ethics approval and consent to participate: **Not applicable**
- Consent for publication: **Not applicable**
- Data availability: **Publicly available**
- Materials availability: **Not applicable**
- Code availability: **Available on Request for reproducibility of results**
- Author contribution: Nilang Bhuvra and Harshvardhan Gupta and Archit Savaliya: contribute by providing insights on data analysis techniques and simulation of deep learning architecture algorithms and wrote the article. Tanmoy Hazra and Saif Nalband provided methodology and proof reading of the article.

References

- [1] Abdelrahman, A., Viriri, S.: Kidney tumor semantic segmentation using deep learning: a survey of state-of-the-art. *Journal of Imaging* **8**(3), 55 (2022)
- [2] Wieclawek, W.: 3d marker-controlled watershed for kidney segmentation in clinical ct exams. *Biomedical Engineering Online* **17**(1), 26 (2018)
- [3] Ronneberger, O., Fischer, P., Brox, T.: U-net: Convolutional networks for biomedical image segmentation. In: *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015. Lecture Notes in Computer Science*, vol. 9351, pp. 234–241. Cham (2015). Springer
- [4] Isensee, F., Jaeger, P.F., Kohl, S.A.A., Petersen, J., Maier-Hein, K.H.: nnu-net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods* **18**(2), 203–211 (2021)
- [5] Causey, A., Starkey, M., Gallagher, J.: An ensemble of u-net models for kidney tumor segmentation with ct images. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* (2021)
- [6] Guo, H., Zhang, X., Zhao, L.: Rau-net: U-net model based on residual and attention for kidney and kidney tumor segmentation. In: *Proceedings of the International Conference on Computing, Electronics and Electrical Engineering (ICCECE)* (2021)
- [7] Geethanjali, B., Dinesh, R.: Semantic segmentation of tumors in kidneys using attention u-net models. In: *Proceedings of the International Conference on Electrical, Electronics, Communication, Computer, and Optimization Techniques (ICEECCOT)* (2021)
- [8] Hou, J., Xiang, T., Wang, S.: A triple-stage self-guided network for kidney tumor segmentation. In: *Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI)* (2020)
- [9] Santini, G., Petersen, A., Sørensen, M.: Kidney tumor segmentation using an ensembling multi-stage deep learning approach. Technical report, KiTS19

- [10] Çiçek, Ö., Abdulkadir, A., Lienkamp, S.S., Brox, T., Ronneberger, O.: 3d u-net: learning dense volumetric segmentation from sparse annotation. In: International Conference on Medical Image Computing and Computer-assisted Intervention, pp. 424–432 (2016). Springer
- [11] Zhao, F., Wu, Y., Chen, Y., al.: Multi-scale supervised 3d u-net for kidneys and kidney tumor segmentation. arXiv preprint arXiv:2005.00567 (2020) [arXiv:2005.00567](https://arxiv.org/abs/2005.00567)
- [12] Milletari, F., Navab, N., Ahmadi, S.-A.: V-net: Fully convolutional neural networks for volumetric medical image segmentation. In: 2016 Fourth International Conference on 3D Vision (3DV), Stanford, CA, USA, pp. 565–571 (2016). IEEE
- [13] Taghanaki, S.A., Zheng, Y., Zhou, S.K., Georgescu, B., Sharma, P., Xu, D., Hamarneh, G.: Combo loss: Handling input and output imbalance in multi-organ segmentation. *Computerized Medical Imaging and Graphics* **75**, 24–33 (2019)
- [14] Heller, N., Isensee, F., Maier-Hein, K.H., al.: The kits23 challenge: Kidney tumor segmentation benchmark. arXiv preprint arXiv:2305.XXXX (2023)
- [15] Cardoso, M.J., al.: Monai: An open-source framework for deep learning in healthcare. arXiv preprint arXiv:2211.02701 (2022) [arXiv:2211.02701](https://arxiv.org/abs/2211.02701)
- [16] The HDF Group: Hierarchical Data Format, Version 5. [Online]. Available: <https://www.hdfgroup.org/HDF5/>. Accessed on: YYYY-MM-DD (1997–2024)
- [17] Neuroimaging Informatics Technology Initiative (NIfTI): NIfTI-1 Data Format. [Online]. Available: <https://nifti.nimh.nih.gov/nifti-1/>. Accessed on: YYYY-MM-DD (2003)
- [18] Khatami, A., Azadeh, P., Ghoroubi, J., Azimi, G.: Adding liver window setting to the standard abdominal ct scan protocol: Is it useful? *Iranian Journal of Radiology* **5**(2), 65–68 (2008)